#1: for task accuracy evaluation, only outputs ~~from~~ a blackbox model and compute the desired metrics.

#2: for perplexity, need LM to generate probabilites and trust that ~~that~~ probabilities sum to 1

The perplexity maxmailst view:

1): Your True distribution is t, model is p
2): Best possible perplexity is $H(t)$, obtained iff $p = t \Rightarrow$ make p close to t
3): if have t, then solve all the tasks
4): so by pushing down on perplexity, will eventual reach AGI.
5): Caveat: this might not be the most efficient way to get there (pushing down on parts of distribution that don't matter)

Caveat: train-test overlap for wikitext splitting

~~Def Knowledge~~ Def knowledge_benchmarks(): #2

Massive Multitask Language Understanding (MMLU):

⇒ 57 subjects: math, history, law ~~...~~ medical, finance, morality
⇒ collected by students from free online sources
⇒ Really about testing knowledge, not language understanding
⇒ Evaluated on GPT-3 using few-shot prompting
⇒ Might cause overfitting/mode saturation

✓ check website: HELM MMLU for visualizing predictions: few shot prompt and predicted answer

⇒ few-shot: format/contents/rank of shots: matters for performance